# 09: Intro to Multiple Regression

Stat 230 - Spring 2026

```
library(ggformula)
library(dplyr)
library(broom)
library(GGally)

supervisor <- read.csv("https://aluby.github.io/stat230/data/supervisor.csv")
#supervisor <- read.csv(here::here("data/supervisor.csv"))
```

> **Note**
>
> This is the quarto version of the R activity. If you prefer to use the tutorial version, you can use https://aluby.github.io/stat230/activities/09-mlr-tutorial

## 1 Data

- In an effort to understand the important aspects of a satisfactory supervisor, clerical employees at a large financial organization were asked to rate their immediate supervisor.
- The survey questions were designed to measure overall performance of the supervisor, as well as six additional characteristics.
- Employees were asked to rate the following statements on a scale from 0 to 100 (0 = "completely disagree", 100 = "completely agree")

*Note:* this data comes from the textbook

| Variable | Description |
|---|---|
| `overall` | Overall rating of supervisor performance |
| `complaints` | Score for "Your supervisor handles employee complaints appropriately." |
| `privileges` | Score for "Your supervisor allows special privileges." |
| `learn` | Score for "Your supervisor provides opportunities to learn new things." |
| `raises` | Score for "Your supervisor bases raises on performance." |

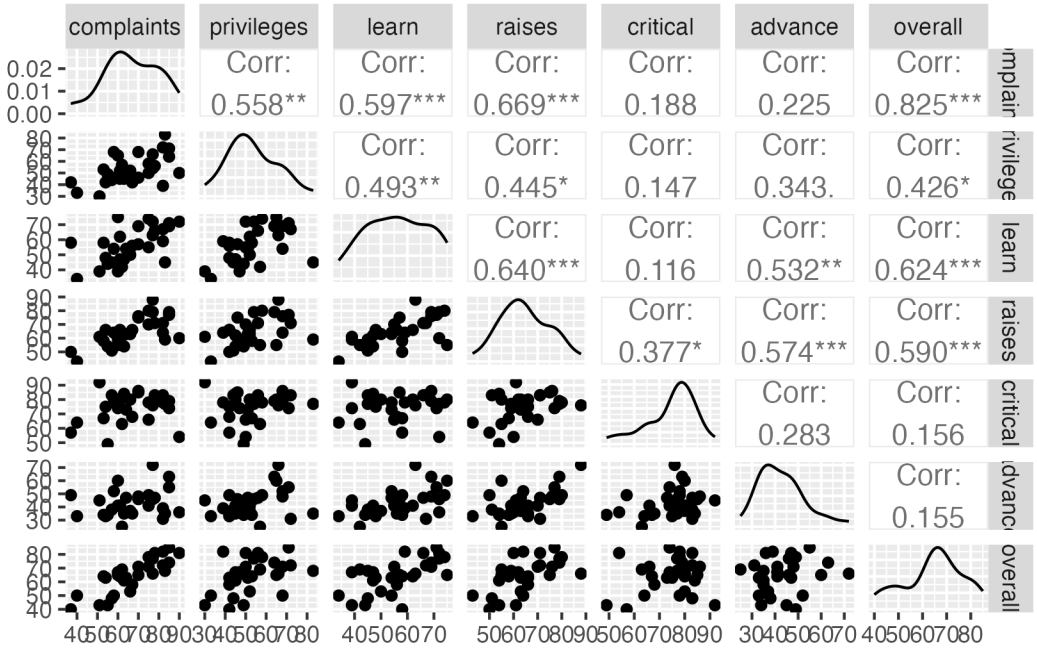
| Variable | Description |
|---|---|
| `critical` | Score for "Your supervisor is too critical of poor performance." |
| `advance` | Score for "I am not satisfied with the rate I am advancing in the company?" |

**Before looking at the data**, how do you think each predictor variable will be related to `overall`? What do you think will have the strongest relationship? The weakest?

## 1.1 EDA: Making a pairs plot

We will use the `ggpairs()` function in the {GGally} package to make the pairs plot. By default, it will include every variable in the dataset. The code below firsts `select()`s the columns we want to include (all of them). I have also moved the response variable to be the last in the list, so that it shows up on the Y axis.

```
supervisor |>
  select(complaints, privileges, learn, raises, critical, advance, overall) |>
  ggpairs()
```




- Which predictor variables appear to be most closely related to `overall`?
- Which predictor variables appear to be most closely related to *each other*?

## 2 SLR

In the code chunk below, choose what you think will be the best *single* predictor and fit an SLR model. Interpret the intercept and slope coefficient and report the $R^2$ value.

```
supervisor_slr <- lm(overall ~ ______, data = supervisor)
summary(supervisor_slr)
```

```
Error in parse(text = input): <text>:1:33: unexpected input
1: supervisor_slr <- lm(overall ~ __
                                   ^
```

## 3 MLR

In the code chunk below, use all predictors and fit an MLR model. Interpret the intercept and two of the slope coefficients and report the $R^2$ value.

```
supervisor_mlr <- lm(overall ~ ______, data = supervisor)
summary(supervisor_mlr)
```

```
Error in parse(text = input): <text>:1:33: unexpected input
1: supervisor_mlr <- lm(overall ~ __
                                   ^
```

## 4 Predicting responses

We can use most our SLR tools for MLR. For example, here's some code I could use to calculate predictions. (Uses the `bikes` data)

```
predict(bikes_lm, newdata = data.frame(temp_actual = 90, windspeed = 6))
```

- Use this as starter code to predict the overall manager score for a manager with scores of 50% for all predictors
- What if the manager from the previous question decides to try to improve her `critical` score by 10 points (so 40% instead of 50%) and her `complaints` score by 10 points (so 60% instead of 50%). How does that change the prediction?

```
# your code here
```

---

## 4.1 Function quick reference

| Function | Purpose |
|---|---|
| `lm(formula, data)` | Fit a MLR model |
| `predict(model, newdata, interval, level, se.fit)` | Make predictions and calculate intervals |
| `data \|> select(vars) \|> ggpairs()` | Make pairs plot for selected variables |